BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 130 · Lesson 130 · Symmetric Encryption: One Shared Secret

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is the main role of a dictionary token in compression?

Q2. What is O(n) search complexity associated with?

Q3. In a BST, which side holds a smaller value?

Q4. In a queue, what does FIFO stand for?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

plaintext — Readable original message.

ciphertext — Encrypted output.

symmetric key — Same secret key is used for encryption and decryption.

key distribution — Securely sharing the secret with authorised parties.

New learning

Start plaintext — Sender has message and shared secret key.

Encrypt — Algorithm combines plaintext and secret key into ciphertext.

Decrypt — Recipient uses matching shared key to restore message.

Worked example

Alice and Bob share secret key K. Explain why intercepting the ciphertext is not enough to read the message.

1. Alice: message + key K — Predict the first step from the prompt.

2. Encrypted ciphertext — Explain the intermediate result or decision.

3. Bob: decrypt with K — A secure symmetric cipher transforms plaintext using key K. Bob decrypts with the same key; an interceptor without K should not recover plaintext feasibly. Security requires a strong cipher and protected key.

We Do · Practise with guidance

A school backs up a sensitive file to a cloud server. Why might symmetric encryption suit bulk data?

Sensitive backup

Predict the first step from the prompt.

Encrypt with shared key

Explain the intermediate result or decision.

Protect key separately

Symmetric algorithms are efficient for encrypting large files, but the encryption key must be stored and shared securely; encrypting a file is not by itself proof of who created it.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Describe the roles of plaintext, key and ciphertext in symmetric encryption. [6]

Question 2

Explain the main key-management challenge in sharing encrypted records between two offices. [6]

Question 3

Evaluate symmetric encryption for a large school backup stored with a third-party provider. [8]

Exit Ticket · Answer before checking

Symmetric encryption uses one public key for encryption and a separate private key for decryption.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. A shorter representation for a repeated substring.

Q2. Linear/sequential search in the worst case.

Q3. The left subtree.

Q4. First in, first out.

You Do pages · Independent questions

Q1. Readable input [2], secret key used by encrypt/decrypt [2], ciphertext unreadable without key [2].

Q2. Both need secret [2], secure initial transfer/storage [2], compromise exposes communications or data [2].

Q3. Bulk speed [2], confidentiality [2], safe key management/recovery [2], practical limitations and conclusion [2].

Exit Ticket · Model answer

False. Those are features of asymmetric encryption.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: